Chapter 3:

Estimation of the Two-Variable Regression Model (OLS)

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Conceptual Summary (Key Points)

1. **Objective of the Ordinary Least Squares (OLS) Method**

To find estimates of β_1 and β_2 that *minimize the sum of squared residuals*:

Minimize $\sum e_i^2$, where:

$$e_i = Y_i - \hat{Y}_i$$

2. **Normal Equations**

Derived by taking derivatives of the SSE with respect to β_1 and β_2:

$$\sum e_i = 0, \quad \sum X_i e_i = 0$$

3. **OLS Estimators**

$$\hat{\beta}_2 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}, \qquad \hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X}$$

4. **Properties of the OLS Regression Line**

- Passes through the means $(\bar{X}, \bar{Y})$.
- The sum of residuals equals zero.
- Residuals are uncorrelated with the explanatory variable X.

5. Classical Linear Regression Model (CLRM) Assumptions

For OLS estimators to be valid and BLUE, these must hold:

- $E(U_i \mid X_i) = 0$ (zero mean of errors)
- $Cov(U_i, U_j) = 0$ (no autocorrelation)
- $Var(U_i) = \sigma^2$ (homoskedasticity)
- $Cov(U_i, X_i) = 0$ (exogeneity)
- Model is correctly specified (no omitted variables or wrong functional form)

6. Properties of OLS Estimators

- **Linearity:** estimates are linear functions of observed data.
- **Unbiasedness:** $E(\hat{\beta}_i) = \beta_i$.
- **Minimum variance among all linear unbiased estimators** (Gauss–Markov theorem → OLS is BLUE).
- **Consistency and asymptotic efficiency** for large samples.

7. Estimating the Error Variance σ^2

Since true errors are unobservable, we use sample residuals:

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2}$$

8. Coefficient of Determination R^2

Measures the proportion of the variation in Y explained by X:

$$R^2 = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$$

Questions + Answers

1. **Why is the sum of residuals zero in OLS?**

Because the first normal equation $\sum e_i = 0$ arises from setting the derivative of the SSE with respect to β_1 equal to zero.

2. **What's the economic interpretation of $\boldsymbol{\beta_2}$?**

It represents the expected change in Y when X increases by one unit, holding other factors constant.

3. **Why is OLS considered the *best linear unbiased estimator*?**

According to the Gauss–Markov theorem, among all linear and unbiased estimators, OLS has the smallest variance.

4. **What happens if the homoskedasticity assumption is violated?**

OLS remains unbiased but loses efficiency — standard errors, t-tests, and F-tests become unreliable.

5. **What's the difference between *unbiasedness* and *consistency*?**

Unbiasedness refers to expected accuracy in finite samples; consistency refers to convergence of the estimator to the true parameter as sample size increases.

6. **Why must the regression line pass through the sample means $(\bar{X}, \bar{Y})$?**

Because by the normal equations, when $X = \bar{X}$, $\hat{Y} = \bar{Y}$.

7. **What happens if $\boldsymbol{Cov(U_i, X_i) \neq 0}$?**

The estimators become biased and inconsistent — OLS is no longer BLUE.

Problem Set (with Answer Guide — Short Solutions)

Problem 3.1 - Computing OLS Estimates

The data below represent weekly income (Y, in $100) and years of schooling (X):

X	Y
10	30
12	38
14	45
16	50
18	59

Tasks:

1. Compute $\bar{X}$ and $\bar{Y}$.
2. Calculate $\hat{\beta}_2 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$.
3. Find $\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X}$.
4. Write the estimated regression line.
5. Predict Y when $X = 20$.

Answer Guide 3.1

$\bar{X} = 14$, $\bar{Y} = 44.4$.

$$\hat{\beta}_2 = \frac{(10-14)(30-44.4)+\ldots+(18-14)(59-44.4)}{\sum(X_i - \bar{X})^2} = 2.9.$$

$\hat{\beta}_1 = 44.4 - 2.9(14) = 3.8$.

→ Estimated line: $\hat{Y} = 3.8 + 2.9X$.

Prediction for $X = 20$: $\hat{Y} = 3.8 + 2.9(20) = 61.8$.

Problem 3.2 – Residuals and Goodness of Fit

Using the estimated line from Problem 3.1, compute residuals $e_i = Y_i - \hat{Y}_i$ for each observation. Then calculate:

1. $\sum e_i$ (should equal approximately 0).
2. $R^2 = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$
3. Interpret the R^2 value.

Answer Guide 2.2

Residuals e_i sum ≈ 0.
$R^2 \approx 0.98 \rightarrow$ 98% of income variation explained by education (excellent fit).

Problem 3.3 – Testing CLRM Assumptions (Conceptual–Quantitative Mix)

For the model $Y_i = \beta_1 + \beta_2 X_i + U_i$, consider:

- $E(U_i \mid X_i) = 0$
- $Var(U_i \mid X_i) = \sigma^2$
- $Cov(U_i, X_i) = 0$

Tasks:

1. Explain what happens if $Var(U_i)$ increases with X_i.
2. Suggest how this can be detected using a residual plot.
3. Explain the impact on OLS efficiency.

Answer Guide 2.3

If $Var(U_i)$ rises with X_i, the model exhibits *heteroskedasticity*.

Residual plot: cone-shaped pattern (variance increasing with X).

OLS remains unbiased but inefficient; standard errors are unreliable.

Problem 3.4 – Variance of the OLS Slope

Suppose the estimated variance of residuals is $\hat{\sigma}^2 = 4$, and the data yield $\sum(X_i - \bar{X})^2 = 50$.

Find $Var(\hat{\beta}_2)$and the standard error of $\hat{\beta}_2$.

$$Var(\hat{\beta}_2) = \frac{\hat{\sigma}^2}{\sum(X_i - \bar{X})^2}$$

Answer Guide 3.4

$Var(\hat{\beta}_2) = 4/50 = 0.08,$

$SE(\hat{\beta}_2) = \sqrt{0.08} = 0.283.$